A Linear Algebra Problem

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This problem comes from "Drinfeld–Sokolov Hierarchies and Diagram Automorphisms of Affine Kac–Moody Algebras".

Conjecture 1. *Let V be a linear space, $v \in V$, $X, Y, Z \in \mathrm{End}(V)$, such that*

$$[X,Y] = Z, \quad [X,Z] = [Y,Z] = 0,$$
$$X^{p+1}v = 0, \quad Y^{q+1}v = 0.$$

Then $(X+Y)^{2p+2q+1}v = 0$.

Consider the algebra $\mathcal{A}$ generated by X, Y with relations $[X,[X,Y]] = [Y,[X,Y]] = 0$. It has a $\mathbb{Z}^2$-grading such that X has degree $(1,0)$ and Y has degree $(0,1)$. Denote the degree-(m,n) part by $\mathcal{A}_{m,n}$. The degree-(m,n) part has basis $Z^i X^{m-i} Y^{n-i}$, $0 \le i \le \min(m,n)$.

For convenience we define $Z = [Y,X]$ from now on.

Lemma 2.

$$Y^n X^m = \sum_{i=0}^{\min(m,n)} C_{m,n}^i \, Z^i X^{m-i} Y^{n-i},$$

where $C_{m,n}^i > 0$.

Proof.

$$\begin{aligned} Y^n X^m &= X Y^n X^{m-1} + [Y^n, X] X^{m-1} \\ &= \sum_i C_{m-1,n}^i Z^i X^{m-i} Y^{n-i} + n Z Y^{n-1} X^{m-1} \\ &= \sum_i C_{m-1,n}^i Z^i X^{m-i} Y^{n-i} + \sum_i n C_{m-1,n-1}^{i-1} Z^i X^{m-i} Y^{n-i} \\ &= \sum_i \left(C_{m-1,n}^i + n C_{m-1,n-1}^{i-1} \right) Z^i X^{m-i} Y^{n-i}. \end{aligned}$$

So each $C_{m,n}^i$ is a sum of positive numbers, hence always positive. □

Remark 3.

$$C_{m,n}^i = i! \binom{m}{i} \binom{n}{i}.$$

Consider the morphism from $\mathcal{A}$ to the Weyl algebra sending $X \mapsto x$, $Y \mapsto \partial$. This morphism is injective when restricted to $\mathcal{A}_{m,n}$. Then one can apply the higher Leibniz rule.

Remark 4. We only need $C^n_{m,n} \neq 0$ when $m \geq n$.

Proof of Conjecture 1. It suffices to prove that if $m+n = 2p+2q+1$ and $M \in \mathcal{A}_{m,n}$, then $Mv = 0$. Without loss of generality assume $n \geq p+q+1$.

Let $M_i = Z^i X^{m-i} Y^{n-i}$ for $0 \leq i \leq m$, and $M'_i = Z^i X^{m-i-p-1} Y^{n-i} X^{p+1}$ for $0 \leq i \leq m-p-1$. Then $M'_i v = 0$, and $M_i v = 0$ if $i \leq n-q-1$.

Denote the span of $\{M'_i\} \cup \{M_i \mid i \leq n-q-1\}$ by W. We need to show $W = \mathcal{A}_{m,n}$, i.e. $M_i \in W$ for all i. Expanding M'_i in terms of M_i, we get

$$M'_i = \sum_{j=0}^{p+1} C^j_{n-i,p+1} M_{i+j}.$$

Note that $m \leq p+q$, so $i \leq m-p-1 \leq q-1$, hence $n-i \geq p+1$, and thus $C^j_{n-i,p+1} \neq 0$.

So we can proceed by induction: assume $M_k \in W$ for all $k < j$, where $j > n-q$. Expanding M'_{j-p-1}, we get a linear combination of $M_{j-p-1}, \ldots, M_j$. Hence $M_j \in W$. And we can reach $j = m$ by expanding M_{m-p-1}. $\square$